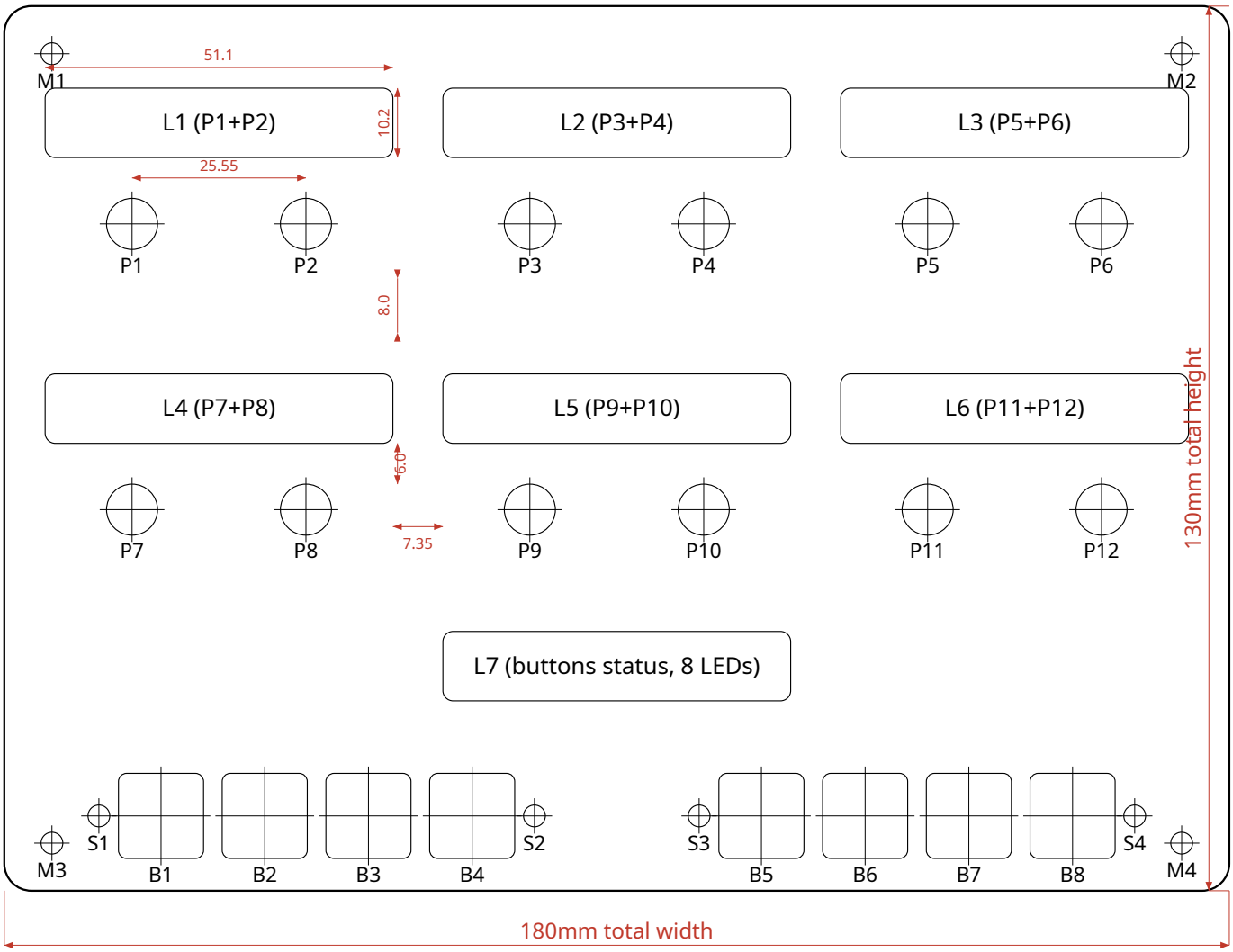
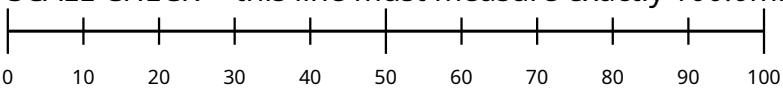


SCALE CHECK -- this line must measure exactly 100.0mm when printed:



PRINT SETTINGS: print at 100% / Actual Size. Disable "Fit to page" / "Scale to fit" / "Shrink oversized pages".
Verify the 100mm scale-check line above with a ruler BEFORE drilling or cutting anything.

Panel: 180 x 130mm. Red numbers are dimension call-outs, shown once near P1/P2/L1 (pot pitch 25.55mm, LED bar 51.1x10.2mm) and once near P7/P8/L4 (pot->LED gap 6.0mm, column gap 7.35mm, row gap 8.0mm) -- all 6 pot pairs / LED bars use identical spacing, so these apply everywhere on the panel.
P1-P12: pot holes, Ø7.5mm -- TODO confirm against your actual pot's bushing thread diameter before drilling.
B1-B8: square cap windows, 12.5x12.5mm (rounded 1.2mm corners), Omron B3F-4055's 12x12mm snap-on cap. Body confirmed via datasheet: 12.5x13.8x7.3mm, PCB/through-hole. v5: pitch tightened 20mm->15.24mm (6x2.54mm grid, perfboard's own hole spacing) so 4 switches/board actually fit a 70mm-long perfboard -- see S1-S4.
S1-S4: Ø3.2mm clearance holes for perfboard mounting screws/standoffs -- S1/S2 flank board 1 (B1-B4) at x=18.9/82.9, S3/S4 flank board 2 (B5-B8) at x=107.1/171.1. Hand-drill these into your own board (70x30mm, centered on each button group, clears the M3/M4 corner posts by ~1.4mm) -- don't rely on its stock corner holes.
v3 change: NL/NR (left/right page-nav buttons, B3F-4055, new in v2) removed -- page navigation is TBD via a different control. L7 no longer shares the row with anything -- it's re-centered alone on the panel's horizontal center (was grouped with NL/NR, off-center, in v2).
L1-L6: LED bar windows, 51.1x10.2mm each -- Adafruit NeoPixel Stick footprint (confirmed). Each sits above its 2 pots, 4 segments/pot.
L7: LED bar window, same footprint, 8 LEDs = 1 per button (status indicator row).
Wiring order: L1=P1+P2, L2=P3+P4, L3=P5+P6, L4=P7+P8, L5=P9+P10, L6=P11+P12 -- chain the NeoPixel sticks in that order (matches README).
v4 change: added a matching 30mm-tall box (see esp-midi-phere-top-template-v5.scad) -- this panel is its lid.
M1-M4: Ø3.2mm clearance holes for the 4 lid-mounting posts, all at the standard 7mm corner inset (v5: M3/M4 moved back to the literal corner -- the pitch shrink above moved B1/B8 clear of it, no special case needed).
v5 open item: the 70x30mm perfboard's Y-extent likely overlaps L7's status-strip footprint in X/Y -- whether they collide depends on mount depth (Z) behind the panel, not decided yet. Plan both before finalizing mounts.
Thin black crosshairs mark hole centers (for center-punch/drill); black circles/rectangles are the cut lines; red lines are reference dimensions, not cut lines.